# LABORATORY SESSION 3: AVR ANALOG-TO-DIGITAL CONVERTER PROGRAMING IN C LANGUAGE

# EMBEDDED SYSTEMS – EC6020

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2019/E/094

GROUP D3

SEMESTER 6

05 JUN 2023

# EXERCISE 01

## C CODE FOR EXERCISE 01

```
#include <avr/io.h>
int main(void) {
    DDRC=0x00; //ADC1 input
    PORTC=0x00; //Uses internal pulldown on PINC
    DDRD=0xff; //D output
    ADMUX = (0 << REFS1) | (1 << REFS0) | (1 << ADLAR) | (0 << MUX3) | (0 <<
MUX2) | (0 << MUX1) | (1 << MUX0);
;
    /*
    REFS1 REFS0
    01   - AVCC as reference, the circuit must contain a capacitor at AREF
    ADLAR
    1    - To keep the 8 most significant bits of conversion in ADCH
    Reserved Bit
    0    - Unused bit
    MUX3 MUX2 MUX1 MUX0
    0001 - Select ADC1 as input for Analog to Digital conversion
    */
    ADCSRA = (1 << ADEN) | (0 << ADSC) | (0 << ADATE) | (0 << ADIF) | (0 <<
ADIE) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0);

    /*
    ADEN
    1    - Enable conversion
    ADSC
    0    - Do not start conversion yet
    ADATE
    0    - No trigger is used
    ADIF
    0    - Interrupt not used
    ADIE
    0    - Interrupt not enabled
    ADPS2 ADPS1 ADPS0
    111 - Select 128 prescale for Analog to Digital conversion clock for
propper conversion
    */
    while(1)
    {
        ADCSRA|=(1<<ADSC); //ADC conversion startup
        while (( ADCSRA&(1<<6))); //Waiting conversion is completed
        PORTD = ADCH; //Sent ADC value to PORT D
    }
}
```

```
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avr-gcc -Os -DF_CPU=16000000UL -mmcu=a
tmega328p -o lab3 lab3.c
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avr-objcopy -O ihex -R .eeprom lab3 la
b3.hex
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avrdude -F -v -c arduino -p atmega328p
 -P /dev/ttyACM0 -b 115200 -U flash:w:lab3.hex

avrdude: Version 6.3-20171130
         Copyright (c) 2000-2005 Brian Dean, http://www.bdmicro.com/
         Copyright (c) 2007-2014 Joerg Wunsch

         System wide configuration file is "/etc/avrdude.conf"
         User configuration file is "/home/eng-co-mpl-013/.avrduderc"
         User configuration file does not exist or is not a regular file, skipping

         Using Port                    : /dev/ttyACM0
         Using Programmer              : arduino
         Overriding Baud Rate          : 115200
         AVR Part                      : ATmega328P
         Chip Erase delay              : 9000 us
         PAGEL                         : PD7
         BS2                           : PC2
         RESET disposition             : dedicated
         RETRY pulse                   : SCK
         serial program mode           : yes
         parallel program mode         : yes
         Timeout                       : 200
         StabDelay                     : 100
         CmdexeDelay                   : 25
         SyncLoops                     : 32
         ByteDelay                     : 0
         PollIndex                     : 3
         PollValue                     : 0x53
         Memory Detail                 :
```

```
                                 Block Poll               Page                       Polled
           Memory Type Mode Delay Size  Indx Paged  Size   Size #Pages MinW  MaxW   ReadBack
           ----------- ---- ----- ----- ---- ------ ------ ---- ------ ----- ----- ---------
           eeprom        65    20     4    0 no       1024    4      0  3600  3600 0xff 0xff
           flash         65     6   128    0 yes     32768  128    256  4500  4500 0xff 0xff
           lfuse          0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           hfuse          0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           efuse          0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           lock           0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           calibration    0     0     0    0 no          1    0      0     0     0 0x00 0x00
           signature      0     0     0    0 no          3    0      0     0     0 0x00 0x00

         Programmer Type : Arduino
         Description     : Arduino
         Hardware Version: 3
         Firmware Version: 4.4
         Vtarget         : 0.3 V
         Varef           : 0.3 V
         Oscillator      : 28.800 kHz
         SCK period      : 3.3 us

avrdude: AVR device initialized and ready to accept instructions

Reading | ################################################## | 100% 0.00s

avrdude: Device signature = 0x1e950f (probably m328p)
avrdude: safemode: lfuse reads as 0
avrdude: safemode: hfuse reads as 0
avrdude: safemode: efuse reads as 0
avrdude: NOTE: "flash" memory has been specified, an erase cycle will be performed
         To disable this feature, specify the -D option.
avrdude: erasing chip
avrdude: reading input file "lab3.hex"
avrdude: input file lab3.hex auto detected as Intel Hex
avrdude: writing flash (178 bytes):

Writing | ################################################## | 100% 0.05s
```

```
avrdude: 178 bytes of flash written
avrdude: verifying flash memory against lab3.hex:
avrdude: load data flash data from input file lab3.hex:
avrdude: input file lab3.hex auto detected as Intel Hex
avrdude: input file lab3.hex contains 178 bytes
avrdude: reading on-chip flash data:

Reading | ################################################## | 100% 0.03s

avrdude: verifying ...
avrdude: 178 bytes of flash verified

avrdude: safemode: lfuse reads as 0
avrdude: safemode: hfuse reads as 0
avrdude: safemode: efuse reads as 0
avrdude: safemode: Fuses OK (E:00, H:00, L:00)

avrdude done.  Thank you.
```

FIGURE 01: COMPILING AND WRITING TO MICROCONTROLLER FOR EXERCISE 01

## RESULTS

LED on => 1

LED off => 0

For 5V all LEDs will on,

So,

Output = $11111111_2$

All LEDs on => $11111111_2$ => 5V

All LEDs off => $00000000_2$ => 0V

Theoretical output for 3.3v => $\frac{255\times3.3}{5} = 168.3$

Practical output for 3.3V => $10011111_2$ => 159

Practical output voltage => $\frac{159\times5}{255} = 3.12V$

# EXERCISE 02

## C CODE FOR EXERCISE 02

```
#include <avr/io.h>
#include <util/delay.h>

#define ADC_THRESHOLD 512

void ADC_Init() {
    ADMUX = (1 << REFS0); // Use AREF as reference voltage
    ADCSRA = (1 << ADEN) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0); //
Enable ADC with prescaler 128
}

uint16_t ADC_Read(uint8_t channel) {
    ADMUX = (ADMUX & 0xF0) | (channel & 0x0F); // Select ADC channel
    ADCSRA |= (1 << ADSC); // Start ADC conversion
    while (ADCSRA & (1 << ADSC)); // Wait for conversion to complete
    return ADC;
}

void LED_Init() {
    DDRD |= (1 << PD3); // Set PD2 (D3) as output
}

void LED_On() {
    PORTD |= (1 << PD3); // Turn on LED (set PD3 high)
}

void LED_Off() {
    PORTD &= ~(1 << PD3); // Turn off LED (set PD3 low)
}

int main(void) {
    ADC_Init();
    LED_Init();

    while (1) {
        uint16_t adcValue = ADC_Read(0); // Read ADC value from LDR connected
to ADC0

        if (adcValue < ADC_THRESHOLD) {
            LED_On();
        } else {
            LED_Off();
        }

        //_delay_ms(100); // Delay for stable readings
```

```
    }

    return 0;
}
```

```
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avr-gcc -Os -DF_CPU=16000000UL -mmcu=a
tmega328p -o LDR LDR.c
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avr-objcopy -O ihex -R .eeprom LDR LDR
.hex
eng-co-mpl-013@engcompl013-Veriton-S2665G:~/Desktop/lab3$ avrdude -F -v -c arduino -p atmega328p
 -P /dev/ttyACM0 -b 115200 -U flash:w:LDR.hex

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         Timeout                       : 200
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         CmdexeDelay                   : 25
         SyncLoops                     : 32
         ByteDelay                     : 0
         PollIndex                     : 3
         PollValue                     : 0x53
         Memory Detail                 :
```

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           hfuse          0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           efuse          0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           lock           0     0     0    0 no          1    0      0  4500  4500 0x00 0x00
           calibration    0     0     0    0 no          1    0      0     0     0 0x00 0x00
           signature      0     0     0    0 no          3    0      0     0     0 0x00 0x00

         Programmer Type : Arduino
         Description     : Arduino
         Hardware Version: 3
         Firmware Version: 4.4
         Vtarget         : 0.3 V
         Varef           : 0.3 V
         Oscillator      : 28.800 kHz
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avrdude: safemode: efuse reads as 0
avrdude: NOTE: "flash" memory has been specified, an erase cycle will be performed
         To disable this feature, specify the -D option.
avrdude: erasing chip
avrdude: reading input file "LDR.hex"
avrdude: input file LDR.hex auto detected as Intel Hex
avrdude: writing flash (250 bytes):

Writing | ################################################## | 100% 0.05s
```

```
Writing | ################################################## | 100% 0.05s

avrdude: 250 bytes of flash written
avrdude: verifying flash memory against LDR.hex:
avrdude: load data flash data from input file LDR.hex:
avrdude: input file LDR.hex auto detected as Intel Hex
avrdude: input file LDR.hex contains 250 bytes
avrdude: reading on-chip flash data:

Reading | ################################################## | 100% 0.03s

avrdude: verifying ...
avrdude: 250 bytes of flash verified

avrdude: safemode: lfuse reads as 0
avrdude: safemode: hfuse reads as 0
avrdude: safemode: efuse reads as 0
avrdude: safemode: Fuses OK (E:00, H:00, L:00)

avrdude done.  Thank you.
```

FIGURE 02: COMPILING AND WRITING TO MICROCONTROLLER FOR EXERCISE 02